

OYYL, Kazakhstan

WMO: 354160

Lat: 49.0656N

Lon: 54.6622E

Elev: 95

StdP: 100.19

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.9	(c) -23.9	(d) -30.3	(e) 0.2	(f) -26.3	(g) -27.4	(h) 0.3	(i) -23.5	(j) 13.7	(k) -6.5	(l) 11.9	(m) -7.4	(n) 2.0	(o) 320	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.2	(c) 37.1	(d) 19.2	(e) 35.4	(f) 18.8	(g) 33.6	(h) 18.3	(i) 20.6	(j) 32.1	(k) 19.9	(l) 31.5	(m) 19.3	(n) 30.6	(o) 3.7	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.7	22.7	16.3	11.8	23.1	15.2	10.9	23.3	59.4	31.6	57.3	31.5	55.2	30.6	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.6	(b) 9.4	(c) 8.1	(e) -29.2	(f) 39.8	(g) 3.7	(h) 1.9	(i) -31.8	(j) 41.1	(k) -34.0	(l) 42.2	(m) -36.1	(n) 43.3	(o) -38.8	(p) 44.7		
(a) 10.6	(b) 9.4	(c) 8.1	(e) -29.3	(f) 22.1	(g) 3.7	(h) 1.0	(i) -32.0	(j) 22.8	(k) -34.1	(l) 23.4	(m) -36.2	(n) 24.0	(o) -38.9	(p) 24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-11.1	-10.6	-1.4	10.0	17.5	22.8	25.2	24.0	16.6	7.6	-0.4	-7.5		
	DBStd	14.12	6.71	7.06	6.13	5.52	4.64	4.47	3.50	4.46	4.73	4.71	5.71	6.42		
	HDD10.0	2622	653	577	354	68	3	0	0	0	6	104	314	543		
	HDD18.3	4464	911	811	611	254	73	12	1	7	87	333	563	801		
	CDD10.0	1828	0	0	2	67	235	384	472	434	204	30	1	0		
	CDD18.3	629	0	0	0	4	47	146	214	183	34	1	0	0		
	CDH23.3	7808	0	0	0	54	576	1837	2649	2283	401	8	0	0		
	CDH26.7	3810	0	0	0	7	213	912	1346	1198	133	1	0	0		
(14) Wind	WSAvg	3.4	3.8	3.8	4.0	3.8	3.6	3.3	3.1	3.0	3.1	3.1	3.2	3.7		
(15) Precipitation	PrecAvg	238	20	17	15	20	21	22	22	16	17	26	26	22		
	PrecMax	353	60	54	35	72	70	68	71	63	59	71	88	75		
	PrecMin	128	2	0	0	3	0	0	0	0	0	0	4	0		
	PrecStd	59	13	13	11	16	18	18	17	15	14	17	16	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.1	17.1	27.7	33.6	38.4	38.8	39.0	33.7	24.2	13.3	5.8		
		MCWB	0.6	1.0	9.5	13.6	17.3	19.1	19.4	19.3	16.2	12.3	9.9	3.1		
	2%	DB	0.7	0.9	13.0	24.8	31.3	36.2	36.7	37.1	30.1	21.5	10.7	2.7		
		MCWB	0.0	0.0	7.6	13.4	17.1	18.9	19.5	19.2	15.0	11.2	7.6	1.5		
	5%	DB	-0.1	0.2	9.5	22.0	29.2	34.1	35.0	35.1	27.7	18.6	8.8	1.3		
		MCWB	-0.6	-0.6	5.7	11.9	16.4	18.3	19.0	18.7	14.4	9.5	6.4	0.2		
	10%	DB	-1.4	-0.9	6.4	19.3	26.7	32.0	33.2	33.1	25.4	15.7	6.7	0.2		
		MCWB	-2.0	-1.7	3.9	10.7	15.3	17.5	18.6	18.1	13.8	8.6	4.8	-0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.2	10.2	15.3	19.5	21.0	21.7	20.9	18.4	14.7	11.5	3.5		
		MCDB	1.6	1.8	15.3	24.2	29.6	32.7	31.6	33.3	29.2	21.6	13.1	4.6		
	2%	WB	0.2	0.3	7.8	14.0	18.2	20.0	20.8	20.1	16.4	12.4	8.5	1.7		
		MCDB	0.7	0.8	12.4	22.1	28.4	32.1	32.0	33.3	25.8	18.0	9.8	2.5		
	5%	WB	-0.7	-0.5	5.9	12.8	17.2	19.2	20.2	19.4	15.3	10.8	6.5	0.4		
		MCDB	-0.1	0.1	9.2	20.5	27.1	31.0	30.9	32.4	25.0	16.6	8.4	1.2		
	10%	WB	-2.0	-1.7	4.0	11.5	16.0	18.5	19.6	18.7	14.4	9.4	4.9	-0.6		
		MCDB	-1.3	-0.9	6.5	18.3	24.9	29.4	29.9	31.0	23.8	14.5	6.6	0.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.7	8.0	11.2	12.0	12.8	12.2	12.9	12.6	10.9	6.5	6.2		
		MCDBR	4.5	5.5	12.0	14.6	15.5	15.4	14.5	15.0	16.1	15.8	9.1	5.3		
		MCWBR	4.6	5.1	7.5	6.3	5.8	5.3	4.6	5.0	6.8	8.3	6.4	4.5		
		MCDBR	4.6	5.0	11.1	12.6	13.8	13.5	12.5	13.4	13.4	13.1	7.6	4.9		
	5% WB	MCWBR	4.7	4.8	7.3	6.0	5.7	5.0	4.4	5.0	6.4	7.8	6.0	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.339	0.400	0.447	0.434	0.423	0.418	0.428	0.377	0.363	0.344	0.319		
	taud		2.075	2.007	2.018	2.024	2.143	2.239	2.266	2.210	2.309	2.263	2.201	2.120		
	Ebn at Noon		695	774	790	796	828	840	838	808	817	752	665	634		
	Edh at Noon		88	120	142	158	146	133	128	130	106	93	79	75		
(44) All-Sky Solar Radiation	RadAvg		1.34	2.50	3.54	4.76	6.11	6.83	6.62	5.74	4.25	2.45	1.19	0.87		
	RadStd		0.21	0.25	0.45	0.44	0.52	0.49	0.37	0.32	0.30	0.29	0.13	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page